

kNN Regression and kNN and Naive Bayes Classification on Baseball Data

Sophia Gavasheli and Jacob Carlson

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Introduction & Research Questions

Our analysis used historical Major League Baseball data from the Lahman Baseball Database

- **Scope:** MLB teams and players in the American League (AL) and National League (NL)
- **Regression RQ:** Predict runs scored by a team by key variables such as at-bats, runs, hits, home runs, stolen bases, walks, and strikeouts, batting average
 - Around 1,500 observations
- **Classification RQ:** Predict a player's position based on batting average, home run average, base on balls average, and strikeout average
 - Around 40,300 observations

Exploratory Data Analysis (EDA)

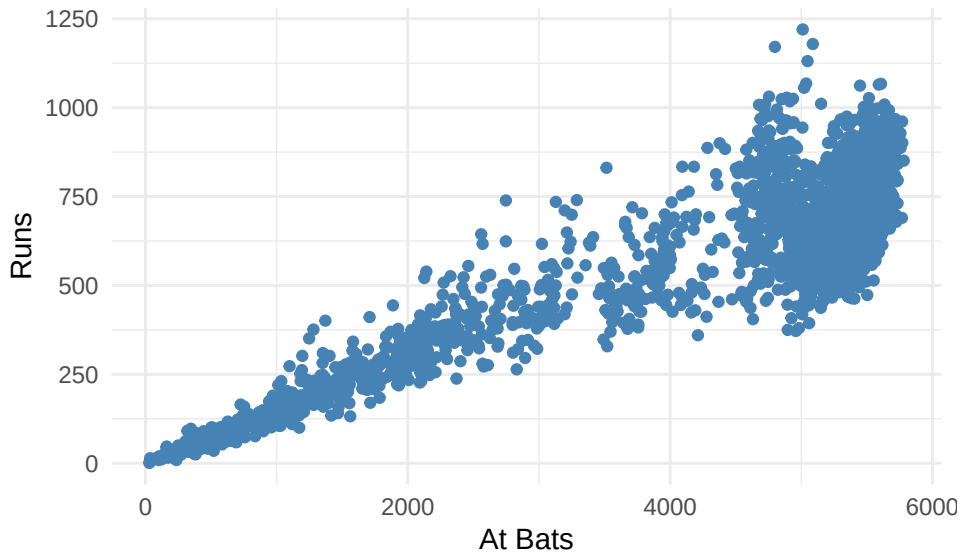
Data Cleaning

- Derived batting average (BA) = Hits / At-Bats
- Player statistics were aggregated by team and year
- Batting statistics were summarized across player positions
- Positions combined into four categories: catcher, infield, outfield, and pitcher

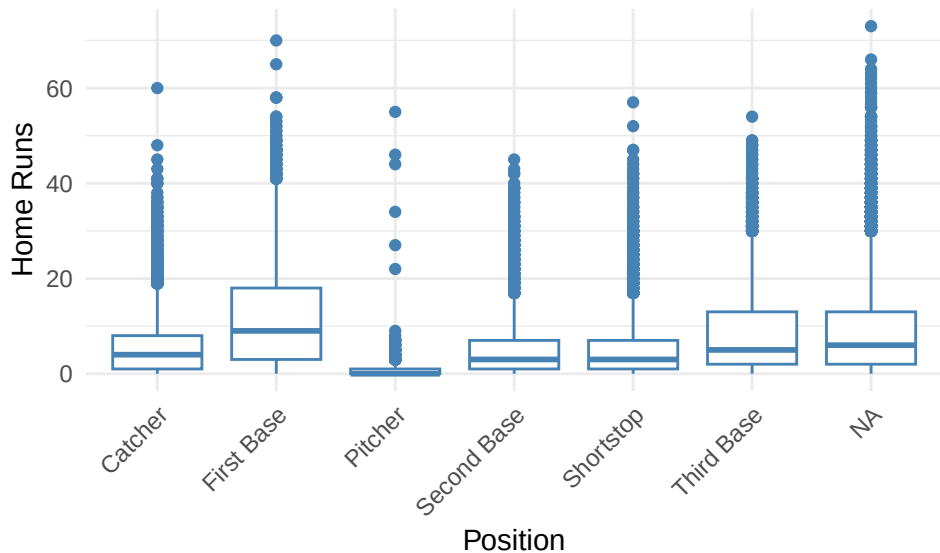
Initial Patterns

- Hits and home runs show a strong positive relationship
- Runs have a relationship with offensive statistics
- Offensive production varies substantially by fielding position

Runs vs At Bats Aggregated by Team



Home Runs by Position



kNN Regression

- Trained on 70% training data scaled with z-scoring
- Optimal k evaluated from $k = 1, \dots, 51$ with cross-validated RMSE which is more sensitive to large errors
- Model performance evaluated on 30% test data with RMSE, MAE, and R^2
- Useful here because baseball statistics may have nonlinear relationships

kNN and Naive Bayes Classification

- Predictors scaled based on at bats
- Multi-class targets: used argmax rule to predict class with largest probability
- Optimal k evaluated with cross-validated balanced accuracy and multi-class log loss
- Model performance evaluated on 30% test data with
 - Log-Loss: penalizes overconfident wrong predictions
 - Brier Score: threshold-free measure of calibration
 - Balanced Accuracy: balances sensitivity and specificity (macro-averaged)
 - Macro-F1: summarizes precision and recall (macro-averaged)

Regression

- Optimal k value was $k^* = 9$
- Model performed better than no-skill model (lower RMSE/MAE, higher R^2)
- The model captures about 82% of the variation in runs per game
 - Offensive stats significantly explain runs scored per game

Table 1: kNN regression on the held-out test set versus the null (mean-only) model.

Model	RMSE	MAE	R-squared
Null (training mean)	0.563	0.450	0.000
kNN ($k^* = 9$)	0.233	0.184	0.829

Results & Key Findings (con't)

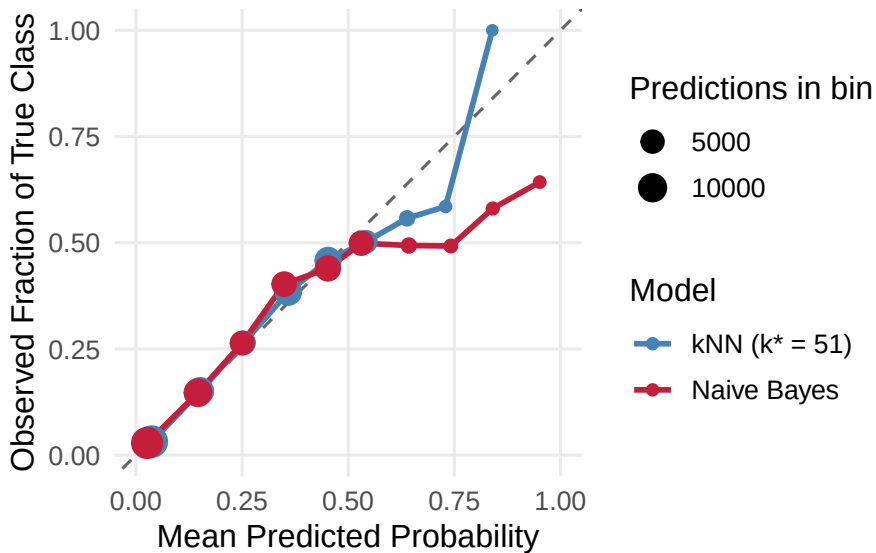
Classification

- Optimal k valued was chosen based on CV log-loss, $k^* = 51$
- Both kNN and Naive Bayes outperform the no-skill baseline
 - kNN produces slightly better probabilistic predictions (lower Log-Loss and Brier Score)
 - Naive Bayes has slightly higher Balanced Accuracy and Macro-F1 (stronger overall class discrimination)
- Which model is best? kNN performs well for probability quality, while Naive Bayes performs well for balanced classification performance.
- Can batting statistics predict player position?
 - Both models perform better than no-skill baseline, so yes.

Table 2: Test-set comparison of the no-skill baseline, kNN, and Naive Bayes.

Metric	No-Skill	kNN ($k^* = 51$)	Naive Bayes
Log-Loss	1.108	1.064	1.069
Brier Score	0.633	0.618	0.622
Balanced Accuracy (Macro)	0.500	0.556	0.563
Macro-F1	0.369	0.345	0.336

Calibration Plot



- The analysis demonstrates that relatively simple machine-learning methods can extract meaningful information from baseball statistics
 - Runs scored per game can be predicted from offensive stats
 - Player positions can be predicted from player performance
- The results also demonstrate why multiple evaluation metrics are necessary: kNN performs better on probabilistic metrics, while Naive Bayes performs slightly better on balanced classification metrics.